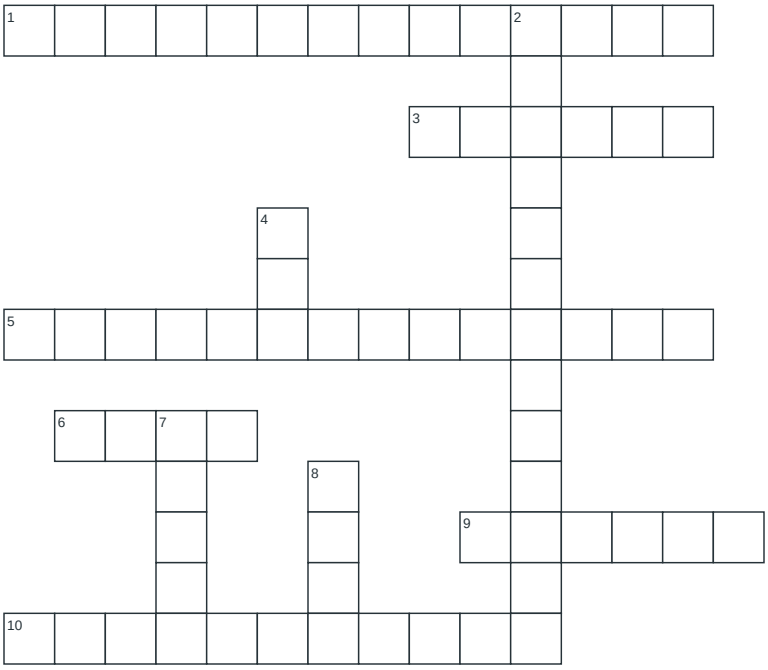
DNA RNA and Protein Synthesis Crossword

Easy | 10 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

1. A supplied key says GCU and GCC both encode alanine. What follows from changing GCU to GCC

3. RNA base that generally replaces thymine

5. Boundary that regulates exchange between a cell and its surroundings.

6. One DNA strand reads ACGT. Which bases pair directly across from it

9. Five-carbon sugar found in RNA

10. Use of an RNA message to assemble a polypeptide

DOWN

2. Synthesis of RNA from a DNA template

4. Molecule that stores hereditary information in most cells

7. Three-nucleotide sequence in messenger RNA that specifies an amino acid or stop

8. RNA molecule that carries an amino acid to a ribosome

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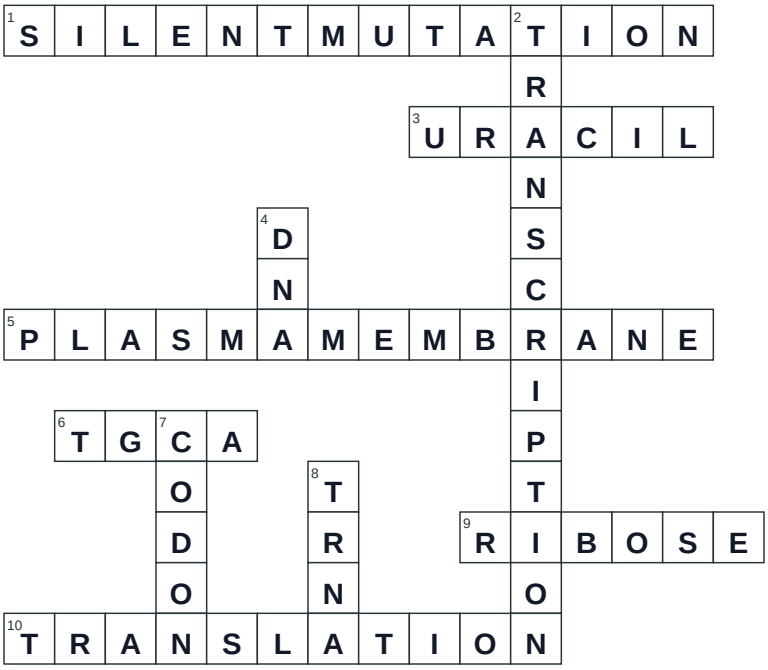
DNA RNA and Protein Synthesis Crossword

Easy | 10 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

1. Silent Mutation
3. Uracil
5. Plasma membrane
6. TGCA
9. Ribose
10. Translation

DOWN

2. Transcription
4. DNA
7. Codon
8. tRNA

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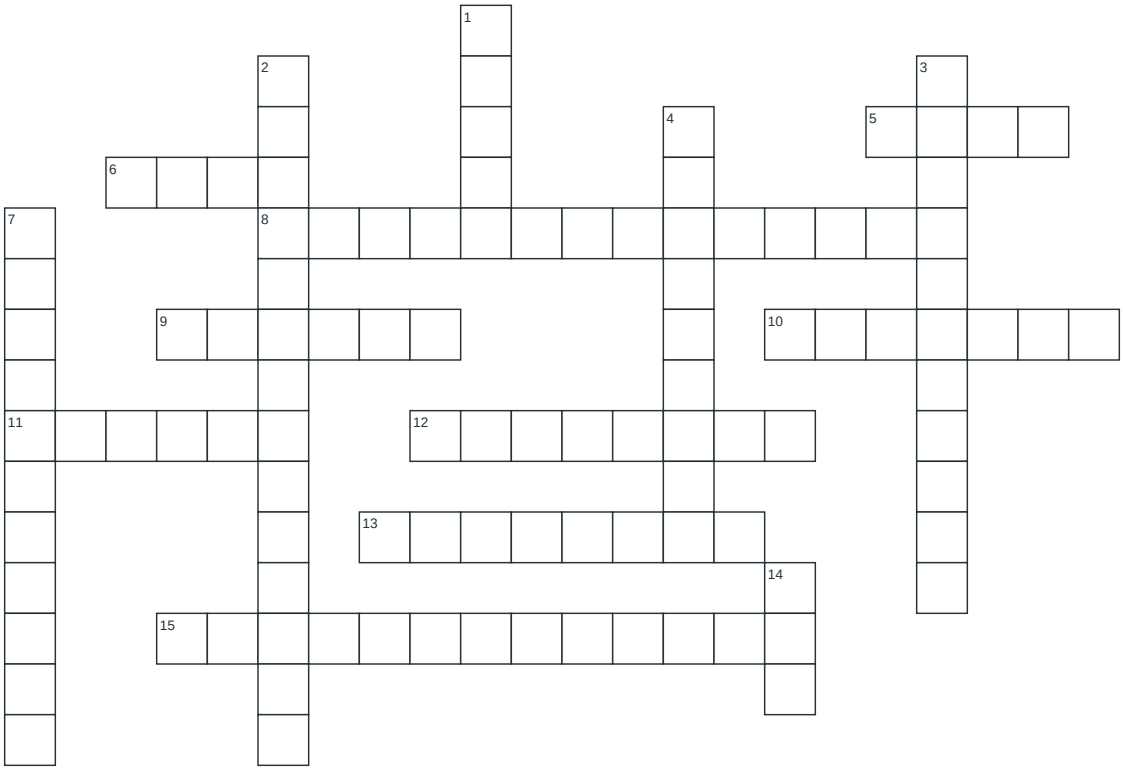
DNA RNA and Protein Synthesis Crossword

Medium | 15 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

5. RNA molecule that carries an amino acid to a ribosome

6. One DNA strand reads ACGT. Which bases pair directly across from it

8. A supplied key says GCU and GCC both encode alanine. What follows from changing GCU to GCC

9. RNA base that generally replaces thymine

10. Membrane-bound organelle that contains most eukaryotic DNA.

11. Five-carbon sugar found in RNA

12. Rigid outer layer that supports many plant, fungal, and bacterial cells.

13. Cell structure that assembles proteins.

ACROSS (continued)

15. Synthesis of RNA from a DNA template

DOWN

1. Three-nucleotide sequence in messenger RNA that specifies an amino acid or stop

2. Boundary that regulates exchange between a cell and its surroundings.

3. Use of an RNA message to assemble a polypeptide

4. Cell contents outside the nucleus and inside the plasma membrane.

7. Organelle where photosynthesis occurs in plants and algae.

14. Molecule that stores hereditary information in most cells

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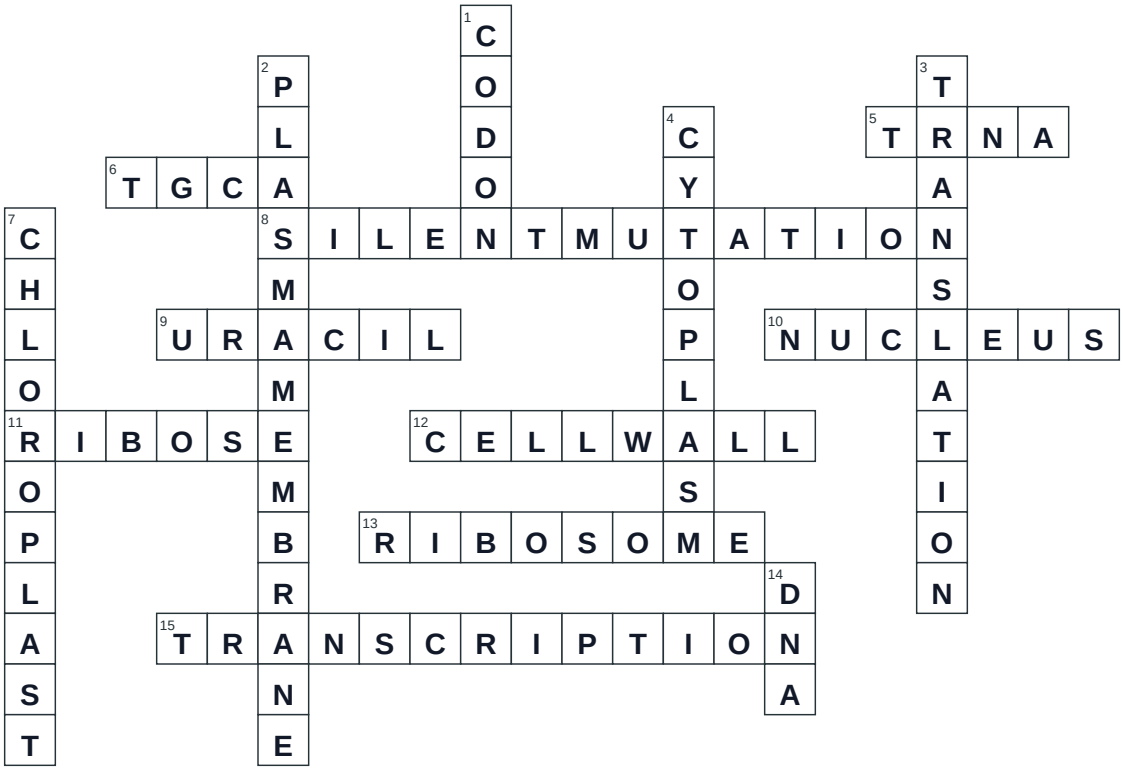
DNA RNA and Protein Synthesis Crossword

Medium | 15 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

5. tRNA
6. TGCA
8. Silent Mutation
9. Uracil
10. Nucleus
11. Ribose
12. Cell wall
13. Ribosome

ACROSS (continued)

15. Transcription

DOWN

1. Codon
2. Plasma membrane
3. Translation
4. Cytoplasm
7. Chloroplast
14. DNA

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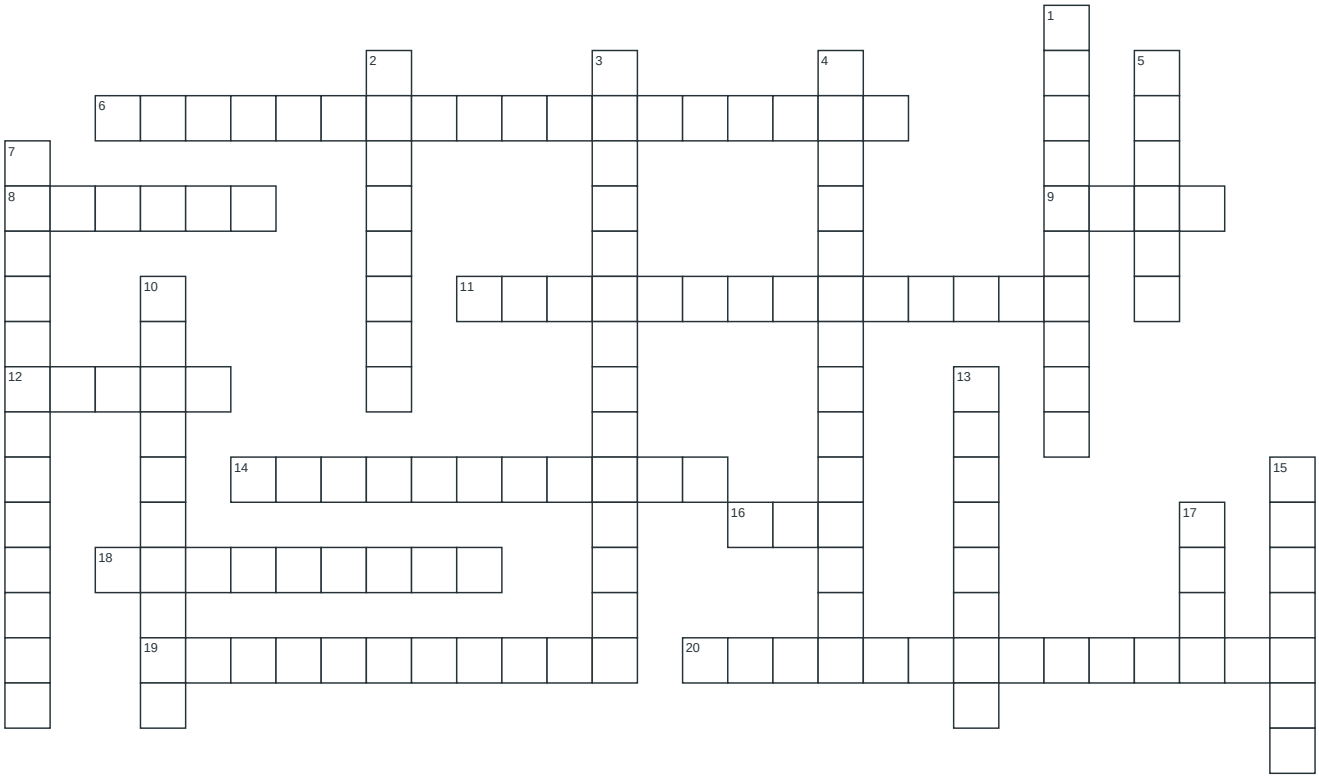
DNA RNA and Protein Synthesis Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

6. Process by which cells develop structures suited to particular functions.
8. Five-carbon sugar found in RNA
9. One DNA strand reads ACGT. Which bases pair directly across from it
11. Plant-cell compartment that stores water and helps maintain pressure.
12. Three-nucleotide sequence in messenger RNA that specifies an amino acid or stop
14. Organelle where photosynthesis occurs in plants and algae.
16. Molecule that stores hereditary information in most cells
18. Cell contents outside the nucleus and inside the plasma membrane.
19. Use of an RNA message to assemble a polypeptide
20. Boundary that regulates exchange between a cell and its surroundings.

DOWN

1. States that living things are made of cells and that cells come from existing cells.
2. Rigid outer layer that supports many plant, fungal, and bacterial cells.
3. A supplied key says GCU and GCC both encode alanine. What follows from changing GCU to GCC
4. Organelle that modifies, sorts, and packages many proteins.
5. RNA base that generally replaces thymine
7. Synthesis of RNA from a DNA template
10. Cell type that lacks a membrane-bound nucleus.
13. Cell structure that assembles proteins.
15. Membrane-bound organelle that contains most eukaryotic DNA.
17. RNA molecule that carries an amino acid to a ribosome

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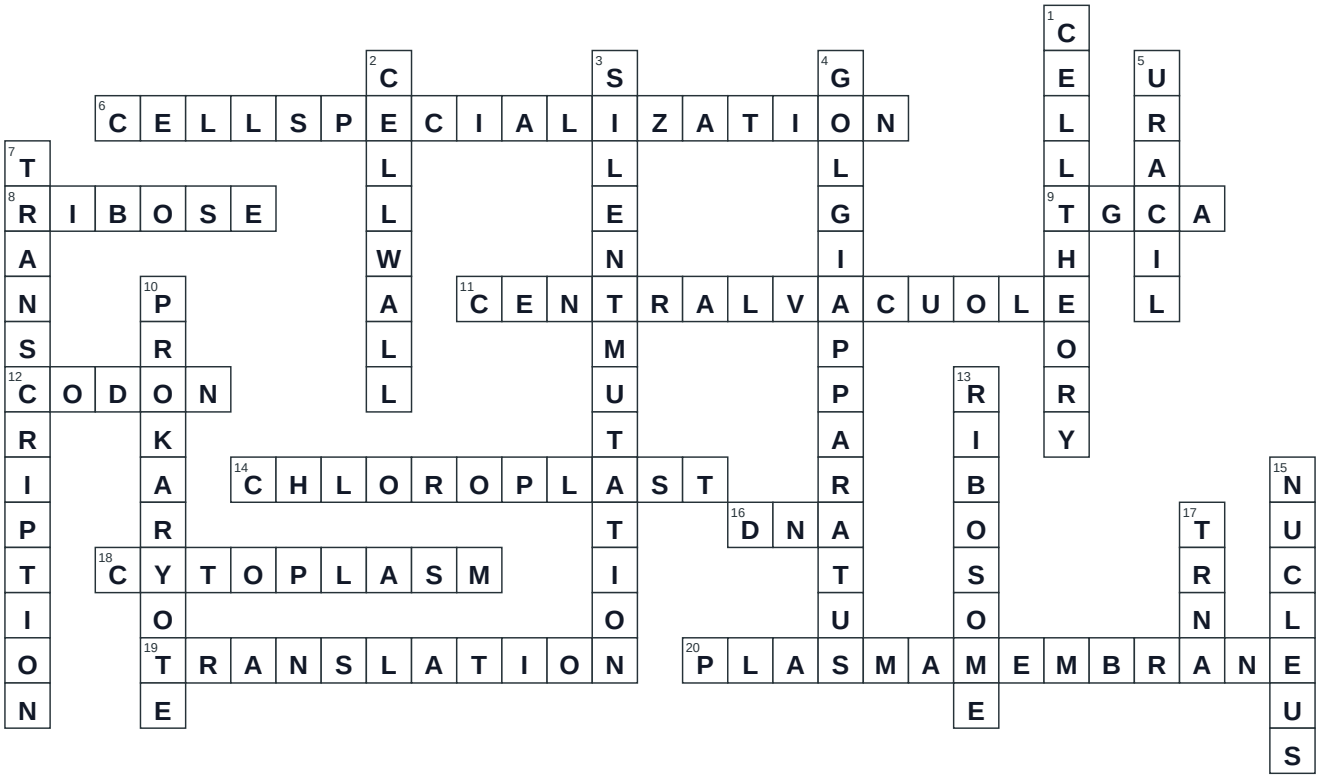
DNA RNA and Protein Synthesis Crossword

Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

6. Cell specialization
8. Ribose
9. TGCA
11. Central vacuole
12. Codon
14. Chloroplast
16. DNA
18. Cytoplasm
19. Translation
20. Plasma membrane

DOWN

1. Cell theory
2. Cell wall
3. Silent Mutation
4. Golgi apparatus
5. Uracil
7. Transcription
10. Prokaryote
13. Ribosome
15. Nucleus
17. tRNA

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